

Intent-Based Autonomous Network Framework Guided by Large Language Model

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Abstract—With the rapid development of next-generation networks, the highly heterogeneous and dynamic nature of networks poses significant challenges for automated network management. Autonomous Network (AN), as a new network paradigm, aims to provide customers with a zero-wait, zero-touch, and zero-fault experience. AN facilitates network management through intent-driven interactions and provides on-demand resource orchestration and service scheduling. However, accurately translating user intents into commands and allocating resources on demand for services remain significant challenges for AN. Therefore, this paper proposes IAN, an intent-based AN framework guided by the Large Language Model (LLM). In the intent translation phase, IAN introduces RAG to enhance command generation quality by retrieving from manuals. In the resource allocation phase, the method utilizes LLM to analyze service characteristics, thereby guiding the training and inference of the resource allocation model to effectively distribute resources uniformly across emerging services. Experimental results demonstrate that IAN improves performance by 52.66% in intent translation tasks and increases overall gain by 33.57% in resource allocation tasks compared to other models.

Note to Practitioners—In this study, the goal is to design an intent-driven autonomous framework. This framework translates user-input natural language intents into network configurations and continuously updates these configurations based on computations of downstream network states to achieve on-demand resource allocation. To accomplish this, we designed two main modules: the Intent Translator and the Resource Allocator. In the Intent Translator, we propose an LLM-empowered intent translation approach. The introduction of RAG enhances the

performance of LLMs in intent translation tasks, effectively mitigating hallucinations and generating higher-quality, deployable commands. In the Resource Allocator, we present an LLM-guided resource allocation method. Leveraging their strong contextual understanding, LLMs can better analyze the attributes of emerging services, thereby guiding the training and inference of resource allocation models. Through this approach, the proposed framework achieves high-quality intent translation and efficient resource allocation, enabling intent-driven network autonomy.

Index Terms—Autonomous network, intent translation, resource allocation, large language model.

I. INTRODUCTION

AS THE new era of connectivity for all things emerges, the highly heterogeneous and dynamic characteristics of networks are becoming increasingly apparent. The advent of new technologies, such as virtual reality and ubiquitous intelligence, is leading to the rapid growth of applications that are offering users entirely new service experiences [1], [2]. In the grand vision for next-generation networks, meeting highly flexible and diverse service needs has become an essential requirement. Additionally, accommodating various application scenarios is crucial for the evolution of future wireless networks [3], [4].

Nevertheless, this transition has made it challenging for operators to efficiently oversee intricate networks through conventional manual techniques. Consequently, there is an urgent need within the industry for the development of automated tools for orchestrating and managing services.

To address this problem, TMForum has proposed a novel network paradigm known as Autonomous Network (AN) [5]. The objective of AN is to develop comprehensive network automation and intelligent operations capabilities. This includes the automatic deployment of user intents and the self-orchestration of services [6]. Efforts within the industry are being made to achieve this goal by combining intent translation with resource allocation. However, accurately translating user intents into commands and allocating resources tailored to specific services are critical for the implementation of AN.

The primary challenges can be summarized into the following three points:

- **Lack of practical solutions for combining semantic-level intent translation with resource allocation.** The process of intent translation is typically conducted at the semantic level. When deploying new commands

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into the network, it is also necessary to allocate resources for these new services. Without this, it can easily lead to resource conflicts and cause network failures.

- **Complex constraints and differences in hardware versions.** The translation of intents into commands is not merely a simple text generation issue; it also involves the consideration of multiple constraints from manuals on command-related aspects, such as syntax and value ranges. Moreover, the functional differences between gNodeB and eNodeB result in some commands being incompatible with each other.
- **Difficulty in modeling different services as quantifiable resource demands.** The continuous emergence of new services presents a challenge in mapping these new services into quantifiable network resource demands, such as bandwidth and maximum latency. This demand ultimately makes it difficult to select an appropriate resource allocation algorithm to guarantee the self-orchestration of services.

The emergence of Large Language Models (LLMs) offers an opportunity to address these challenges. Relying on LLM's excellent text generation capabilities, they are exceptionally suited for intent translation tasks. Furthermore, LLM models demonstrate a high degree of logical understanding, which enables them to model emerging services in terms of specific Quality of Service (QoS) metrics. This capability serves to inform the training and decision-making processes associated with resource allocation models [7], [8].

We propose IAN, an intent-driven autonomous framework guided by LLM. IAN is comprised of two primary components: the Intent Translator and the Resource Allocator. The Intent Translator converts user-input natural language intents into deployable commands. To ensure the effectiveness of intent translation, we introduce Retrieval-Augmented Generation (RAG) to enhance the quality of the commands generated by the Intent Translator. Once the new commands are deployed into the network, the Resource Allocator first identifies the services contained within the intents using LLM. It then guides a deep reinforcement learning (DRL) model to allocate resources to the corresponding services, thereby achieving an intent-based autonomous feedback loop. The main contributions of this paper are as follows:

- 1) We introduce IAN, an intent-based AN framework guided by LLM. Through the guidance of LLM, we achieve network autonomy and on-demand services by directing both intent translation and resource allocation.
- 2) We propose an LLM-empowered intent translation method. The incorporation of RAG serves to enhance the performance of LLMs in intent translation tasks, effectively mitigating the illusion of LLMs and generating higher-quality deployable commands.
- 3) We present an LLM-guided resource allocation method. With their strong ability to capture context, LLM can better analyze the attributes of new services when they emerge, thereby guiding the training and inference of resource allocation models.

II. RELATED WORK

A. Intent-Based Autonomous Network

The traditional approach to network management is based on the expertise of human professionals and automated scripts for monitoring and adjustment. However, with the rapid changes in demand and increased deployment costs, this approach is no longer sufficient. Additionally, the growing difficulty of retraining network administrators exacerbates the challenge of meeting current network needs [9]. AN management, as a potential solution, has quickly attracted significant attention from the network community following the proposal of software-defined network (SDN). It aims to address these challenges, improve network and service provisioning processes, and meet the needs of different users and service providers [10].

The advent of network technology has led to the suggestion that intent could serve as an interface between operators and the network. An intent-based AN is an intelligent network system that can autonomously transform, deploy, and configure network resources in accordance with the operator's intent. Lumi [11] employs the abstract language Nile [12] to interpret the operator's intent and subsequently transforms it into a policy. Szyrkowiec et al. [13] proposed an architecture that can automatically provide secure services at multiple levels (IP, Ethernet, and optical networks) based on intent. Concurrently, it uses an open-source SDN orchestrator to select the appropriate encryption layer. This architecture achieves automated management across various network layers through intents and provides some regulation of network resources. However, intent translation is performed solely through template matching to generate automated actions at each layer, and the regulation of network resources lacks a connection to the intent. Reference [14] designed a natural language-based intent translation system that leverages LLMs to convert users' network requirements expressed in natural language into network service descriptors deployable on edge computing clusters. Additionally, to enhance the system's accuracy, a human feedback loop mechanism was introduced into it. However, the system utilizes LLMs only for entity extraction to generate fixed templates, which does not fully leverage the understanding capabilities of LLMs. Additionally, the introduction of human feedback reduces the level of automation. CoSynth [15] tried to create configuration commands using GPT-4 [16] and then fix them with multi-layer validators. However, these methods separate domain knowledge from network dynamics.

Previous studies have primarily investigated intent translation and resource allocation scheme generation in isolation, leading to a highly modular AN architecture. Intent translation typically necessitates alignment with the underlying network infrastructure. Otherwise, it may result in significant configuration mismatches. Additionally, resource allocation schemes for services must be based on a comprehensive understanding of service requirements to avoid service failures due to insufficient resources. By leveraging LLMs for understanding and perceiving intents and services, IAN applies these capabilities to intent translation and

resource allocation tasks, thus realizing intent-driven network autonomy.

B. Large Language Model

In the field of Natural Language Processing (NLP), LLMs such as ChatGPT have brought about revolutionary changes. Notable LLMs in the English domain include GPT-4 and Llama2 [17], while ChatGLM [18] and ERNIE [19] are prominent examples in the Chinese domain. Additionally, DeepSeek [20] and Qwen-3 [21] have also emerged as powerful LLMs in 2025, contributing significantly to the advancements in the field. Characterized by their extensive scale based on the transformer architecture, these LLMs are trained on a wide array of text datasets with billions of parameters. This significantly enhances their ability to understand and generate complex, fine-grained language [22]. Nevertheless, the utilization of pre-trained models is inadequate for the specific or out-of-scope conversational tasks. Consequently, techniques such as supervised fine-tuning are extensively employed. Supervised fine-tuning entails the training of the model on labeled datasets, wherein human annotators furnish examples of the desired input-output behavior [23]. This methodology enables pre-trained LLMs to learn task-specific abilities and perform more effectively in scenarios requiring language understanding, generation, and interaction [24], [25]. As evidenced by Xu et al. [26] and Guo et al. [27], supervised fine-tuning techniques are used to enhance the language translation capabilities of LLMs.

C. Retrieval-Augmented Generation

To address issues such as slow domain knowledge updates, the need for improved data accuracy, and the generation of hallucinations, RAG method has been proposed as a potential solution [28], [29], [30]. The RAG method markedly enhances the precision of LLM responses, while simultaneously ensuring real-time data updates and preventing the misuse and loss of data [31].

PET-SQL [32] proposes a prompt-enhanced two-round text-to-SQL framework that optimizes schema linking through preliminary SQL generation and introduces cross-model consistency validation. TPTU-v2 [33] presents an integrated framework incorporating an API retriever, LLM finetuner, and adaptive demo selector to address challenges in large-scale API invocation, task planning ambiguities, and distinguishing similar tools in complex real-world systems, significantly enhancing LLM agents' task execution capabilities. A pivotal advancement is Agentic RAG, which embeds autonomous agents into the RAG pipeline to enable iterative refinement, adaptive retrieval strategies, and multi-agent collaboration [34]. These agents leverage design patterns like reflection, planning, tool use, and hierarchical workflows to dynamically optimize retrieval and generation. For instance, systems like Corrective RAG [35] and Adaptive RAG [36] dynamically evaluate query complexity, validate retrieved documents, and refine outputs through feedback loops, excelling in healthcare diagnostics, financial analytics, and customer support.

Intent	#	Command Name	Parameter Name	Parameter Value
“Ensure that users can perform 4K video conferencing stably.”	1	MOD CELL	LocalCellId	1
			BandWidth	BW_N100
	2	MOD CELLALGOSWITCH	LocalCellId	1
			RoheSwitch	ON
			ServiceDiffSwitch	ON
	3	ADD ASPARAGROUP	AsParaGroupID	2
			AsPreallocDuration	2000

Fig. 1. The examples of intent and commands for ensuring user video conference service.

D. DRL-Enabled Resource Allocation

Network resource allocation is a common consideration in network environments. Mathematical models based on traditional approaches face significant challenges, including high computational complexity, lack of adaptability, and robustness [37]. RL is similarly constrained by limitations in the state space and its inability to handle complex tasks. Moreover, there is a need to enhance the generalization ability of RL models. Advanced DRL methods are now widely used in the field of resource allocation [38]. Given the diverse effects that resource allocation can have on the physical layer state, the network resource scheduling problem is typically formulated as a Markov Decision Process (MDP) problem and solved by DRL methods [39]. Han et al. [40] employed the DuelingDQN algorithm to facilitate the learning of resource allocation strategies, including the deployment of Virtual Network Functions (VNFs) and the distribution of network bandwidth. Dong et al. [41] proposed a graph convolutional network-driven multi-task DRL to handle network slicing resource allocation and routing selection problems in multi-user scenarios. These successful applications further illustrate the efficient practicality of DRL in network resource allocation.

III. PROBLEM DESCRIPTION

The intent translation function in AN translates user intents into multiple deployable commands. As shown in the Fig 1, this is a data example of intent translation. On the left is the user intent, while on the right are the commands for implementing the intent. Each command includes several parameters and their values. The figure only displays a portion of the commands and related parameters; generally, each command contains 2–18 parameters. The user's intent is: “Optimize the stability of 5G users' live streaming.” To achieve this intent, 1, 2, and 3 commands need to be sent to the controller. 1 modify the resource allocation configuration of the cell. 2 adjusts the algorithm switch of the cell. 3 added configurations to ensure the quality of user service experience.

The deployable commands must adhere to the syntax specified in the manual, and the manual strictly regulates the value of each parameter. The manual includes the command name, a description of the command, the parameters involved, descriptions of each parameter, and the range of values for each parameter. Therefore, the generation of high-quality

commands under the intricate constraints of the manual represents a considerable challenge, which constrains the ability to translate intent effectively.

In order to achieve resource allocation and make the deployed command more reasonable, it is essential to take into account the status of the downstream network for computation, including factors such as bandwidth. The nodes in the network are composed of user terminal nodes, wireless base stations, and switches. The type of base station considered is mainly macro base station equipment.

We take into consideration a downstream network topology $G = \{N, L\}$, with N nodes and L links. Each link l ($l \in L$) is allocated a specific bandwidth resource B_l . There are multiple user services U in the network. Each user can generate multiple services. For each user service u , there is a source node n_u and a destination node d_u . The set L_{n_u, d_u} represents all the links along the route connecting these two nodes. Let $O = \{o_1, o_2, \dots, o_m\}$ ($m \leq |O|$) represent the various QoS indicators contained in the user service u ($u \in U$). The network must receive the user's intent and subsequently convert it into a weight matrix $W_u = \{w_{u, o_1}, w_{u, o_2}, \dots, w_{u, o_m}\}$ through analysis. This is done to measure the importance of the various QoS in the service.

The network gets rewards based on the overall Service Level Agreement (SLA) Satisfaction Ratio (SSR) and the total benefits of QoS by allocating bandwidth B_u to fulfill all user services. In order to compute the SSR, a binary variable, denoted by $Y_u \in \{0, 1\}$, is introduced to indicate whether the bandwidth currently allocated meets the requirements of the QoS indicators for the service.

The allocation of bandwidth resources can cause a certain amount of link resource loss, potentially leading to a decrease in network performance, such as increased latency or packet loss. This loss will incrementally increase as the link bandwidth resources become strained. Given these factors, we define the method by which the network calculates the benefit of each QoS indicator o_m in each service u requirement:

$$R_{u, o_m} = \beta_1 F(O_m(B_u, S_u)) - \beta_2 \sum_{l \in L_{n_u, d_u}} \frac{B_u}{B_l} \quad (1)$$

where β_1 signifies the payment received by the network from the user. $O_m(\cdot)$ is the method of calculating the QoS indicator o_m . S_u represents the status of the resources traversed by the user to the server. β_2 is the cost coefficient, reflecting the network's link loss when allocating bandwidth. The maximum value of the QoS indicator o_m that the current network can provide to the user is defined as $o_{m, max}$. In contrast, the minimum value of the QoS that the service can accept is defined as $o_{m, min}$. The utility function, denoted by $F(\cdot)$, calculates the current QoS metric value based on the allocated bandwidth and normalizes this metric to determine the degree to which the service's QoS requirements are met. This normalized value is then used to assist in calculating the service fee that the respective user needs to pay. Therefore, $F(\cdot)$ can be calculated as follows:

$$F(O_m(B_u, S_u)) = \frac{O_m(B_u, S_u) - o_{m, min}}{o_{m, max} - o_{m, min}} \quad (2)$$

By combining the QoS weight allocation matrix W_u , we can derive the total QoS benefits for each service u :

$$R_u = \sum_{o_m \in O} w_{u, o_m} \cdot R_{u, o_m} \quad (3)$$

In summary, we define the optimization function for the network as follows:

$$\max_{B_u} \quad \alpha_1 SSR \cdot r + \alpha_2 \sum_{u \in U} R_u \quad (4a)$$

$$\text{s.t.} \quad B_u \leq B_l, \forall l \in L_{n_u, d_u} \quad (4b)$$

$$B_u \in B_a \quad (4c)$$

where α_1 and α_2 are coefficients that represent the significance of the rewards from SSR and the benefits from QoS, respectively. r indicates the additional unit service fee obtained by the network from the user based on the SSR fulfillment. $B_a = \{b_1, b_2, \dots, b_a\}$ represents the a available settings for bandwidth value. Eq. (4b) aims to prevent the allocated bandwidth from exceeding the remaining resources on each link. Eq. (4c) defines the acceptable range for optional bandwidth parameters. The network needs to ensure that users are allocated an appropriate bandwidth to avoid network resource conflicts between users and to maximize the overall benefits of the network.

IV. METHOD

A. Overview

The main process of IAN, as shown in Fig. 2, comprises two phases: the intent translation phase and the resource allocation phase. During the intent translation phase, the Intent Translator translates user intents into deployable commands while simultaneously optimizing the command content based on the retrieval results from the external knowledge base (KB). In the resource allocation phase, Resource Allocator analyzes user intents in conjunction with the network state to allocate resources to users, and modifies the corresponding resource parameters in the commands. The Intent Translator consists of the Intent Translation LLM (ITLLM), the Retrieval-Enhanced Module, and the KB, whereas the Resource Allocator is constituted of Service Analysis LLM (SALLM) and the DRL model.

IAN utilizes LLM to guide an intent-based autonomous framework. The overall framework integrates intent translation with resource allocation, translating intents into deployable commands, and orchestrating resources uniformly for services. The integration of LLMs with RAG enables the translation of intents under the complex constraints of the manual, thereby significantly enhancing the quality of the generated commands. The specific steps are as follows: 1) The retrieval-enhanced module combines intent with a two-stage hybrid retrieval algorithm to search for external knowledge. 2) ITLLM generates deployable commands by combining intent and external knowledge. 3) The resource allocator uses SALLM to analyze the weight of QoS indicators. 4) The DRL model allocates corresponding resources for the service based on network conditions and modifies the corresponding parameter values in the deployable commands.

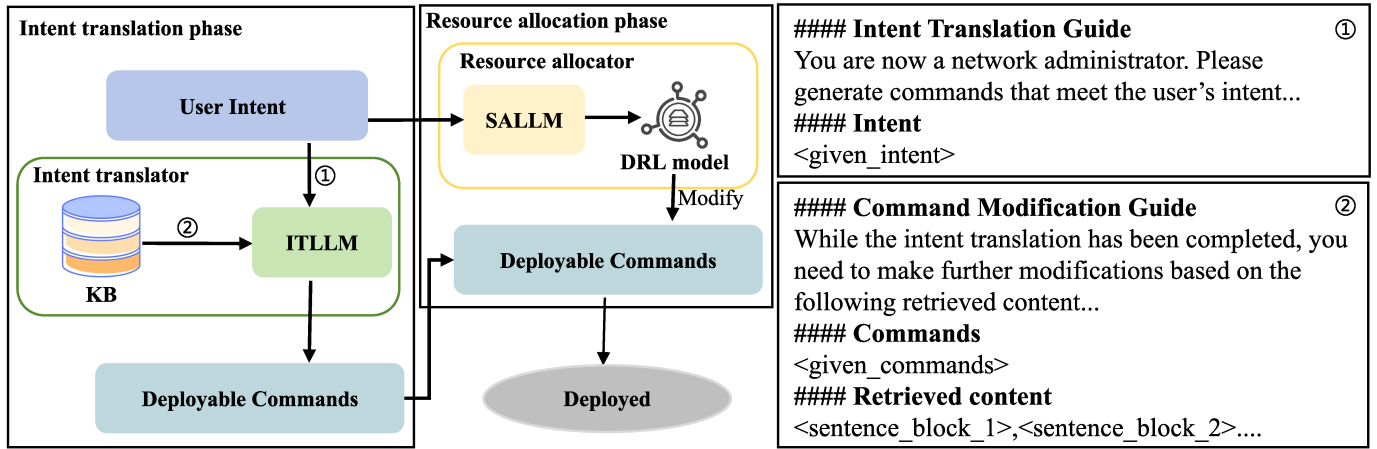


Fig. 2. The workflow of IAN involves two phases. During the intent translation phase, the system generates specific commands that can be sent to network devices based on user intents and information retrieved from the KB. Examples of prompts used in the intent translation phase are provided on the right side of the figure. In the resource allocation phase, the DRL model utilizes the weights of QoS indicators analyzed by ITLLM to generate an optimal resource allocation scheme and updates the executable commands in real time.

B. Intent Translator

Intent serves as a natural language carrier expressing user needs and is an effective way to manage networks. Users control networks and create new services through intents in AN. However, the diversity of natural language expressions often makes it challenging for the intent translation interface to convert complex intents into actionable network configurations. The emergence of LLMs has significantly alleviated this issue. The essence of the intent translation task is a collection of various language tasks, such as entity extraction and text generation. LLMs, having been pre-trained in large-corpus datasets, exhibit robust multitasking capabilities and contextual understanding. Moreover, device manuals serve as a crucial corpus, providing extensive descriptions of commands and restrictions on parameters, which can assist LLMs in the intent translation task. Consequently, we propose an intent translation algorithm guided by LLMs, which incorporates the RAG with a two-stage hybrid retrieval algorithm to enhance the LLM's command generation capabilities.

The core idea of RAG is to combine retrieval and generation, enhancing the accuracy and relevance of the model's output. Traditional RAG retrieves relevant information from KB and inputs this information as context into the LLM to assist in generating responses. This method improves the efficiency of accessing network knowledge, but when dealing with complex user intents, it may still be affected by information omission or inadequate semantic matching. To address this, we introduce a two-stage retriever to more precisely match user intent and deployable commands in the output results. In the first stage, the system employs weighted keyword retrieval to filter out document sets related to the user intents quickly; in the second stage, vector retrieval is used to convert the filtered documents into vector representations, calculating the semantic similarity between the documents and the user intents, further refining the matching results. Through this dual-retrieval strategy, the two-stage retriever can avoid the information loss caused by relying solely on keyword retrieval and improve the accuracy of semantic matching, ensuring

that the returned documents align with the user intents both in terms of keywords and semantics. Ultimately, this new mechanism provides more precise context to the LLM, helping it generate deployable commands that better meet actual user needs [31].

The Intent Translator module is the primary component responsible for intent translation in IAN, consisting of three main components: ITLLM, Retrieval-Enhanced Module, and KB. ITLLM is a LLM fine-tuned by manual corpus and intent translation dataset. The retrieval-enhanced module uses keyword weighting and a multi-stage retrieval mechanism to accurately match and integrate text blocks related to the intent and original deployable commands from the KB, ultimately providing the document with the highest overall score. GPT-4 is employed to divide the manual into discrete text segments, which are then assembled into a KB. As shown in Fig. 3, the Intent Translator primarily involves the following three steps:

- **Step 1: Offline fine-tuning training of ITLLM.** ITLLM is generated by fine-tuning the ChatGLM-6B model based on the dataset. The dataset includes a part that consists of manufactured intent translation datasets. Another part is derived from the command and parameter-related corpus parsed by GPT-4 from the manual. There are four datasets used for fine-tuning ITLLM, and the details are shown in Table I.
- **Step 2: Online intent translation with ITLLM.** Through fine-tuning, ITLLM has developed preliminary intent translation capabilities, allowing it to generate initial deployable commands based on user intent. ITLLM is then deployed online to perform intent translation tasks.
- **Step 3: Online modification of the deployable commands by RAG.** The intent and commands generated in the previous step are used as input to the Retriever query, which then retrieves the top-k-related text blocks. These text blocks, along with the intent and the deployable commands, serve as prompts for input into ITLLM. ITLLM adjusts the parameters and their values within the

TABLE I
DETAILS OF THE FINE-TUNING DATASET FOR ITLLM

Dataset Name	Input Example	Output Example	Source
Policy name and description	What is the role of ACT CELL?	This command is used to activate the cell.	Manual
Parameter and description	What does CellRadius mean?	This parameter represents the radius of the cell...	Manual
Policy to parameter mapping	What are the parameters contained in the instruction ADD SCSSERVICEQOS?	The ADD SCSSERVICEQOS command contains the following parameters: ScQosId,...	Manual
Intent-command sequence mapping	Ensure that users can perform 4K video conferencing stably.	MOD CELL: LocalCellId=0,BandWidth=...	Man-made

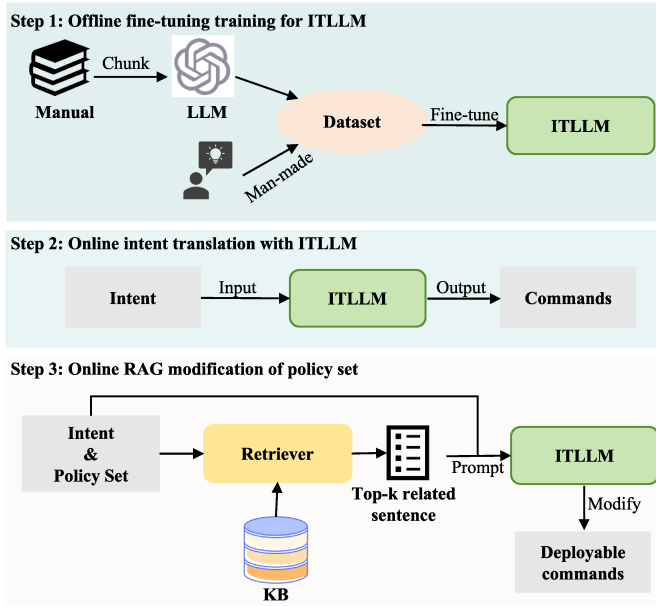


Fig. 3. The workflow of the Intent Translator involves three steps: Step 1 involves the offline fine-tuning of ITLLM; Step 2 involves the online inference with ITLLM; and Step 3 includes the introduction of RAG for online modification of the deployable commands.

deployable commands based on the content of the new prompt, ensuring that the deployable commands better align with the intent requirements and conform to the deployable commands standards.

The LLM-based intent translation algorithm fully leverages the excellent language comprehension and generation capabilities of LLMs, enhancing the accuracy of intent translation tasks. Additionally, the manual serves as a primary source of external knowledge, enhancing the LLM's ability to generate deployable commands through retrieval.

C. Two-Stage Hybrid Retriever

The Retrieval-Enhanced Module is built based on two-stage hybrid retriever algorithm. The algorithm significantly

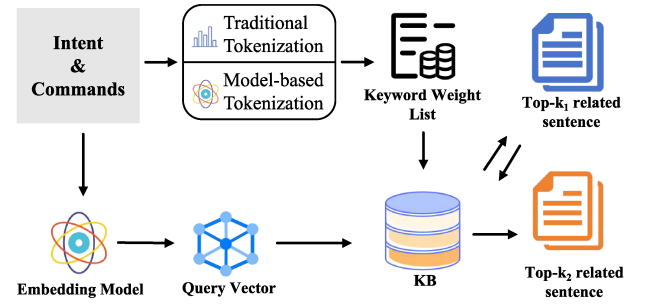


Fig. 4. The two-stage retriever is divided into two stages. The first stage uses a keyword retrieval algorithm based on weighted factors, while the second stage uses vector retrieval. This module improves the retrieval capability of intent and deployable commands within the KB.

enhances the accuracy of knowledge retrieval in the RAG technique, effectively preventing information omission or query bias. As shown in Fig. 4, two-stage retriever first combines the user intent and deployable commands with a keyword retrieval algorithm based on weighted factors. This step allows for rapid filtering of document sets containing specific network metric keywords from the KB. Then, a deep learning-based semantic matching model is introduced for vector retrieval. It calculates the similarity between the user intent vector, deployable commands vector, and the entity vectors in the KB, further refining the selection results. This step ensures that the final returned documents not only contain the keywords, but also have high semantic relevance to the user intent.

This hybrid retrieval strategy not only improves the retrieval speed in network service scenarios, but also effectively avoids the information omission caused by pure keyword matching or the query bias that may arise from relying solely on semantic matching. As a result, it achieves a more intelligent and efficient knowledge retrieval experience, providing users with timely and accurate service responses. The detailed two-stage hybrid retrieval algorithm is elaborated in Algorithm 1.

1) *Keyword Retrieval Stage*: The two-stage retriever takes the user intent and the deployable commands obtained from the ITLLM model. It extracts keywords from the input using a keyword weight model and assigns a corresponding weight

Algorithm 1 Two-Stage Hybrid Retrieval Algorithm

Input: User query Q , deployable commands P , KB $\mathcal{KB}$, Keyword weight model M

Output: Top- k_2 relevant documents $\mathcal{D}_{top}$

```

1:  $\mathcal{K}_Q, \mathcal{K}_P \leftarrow \text{ExtractKeywords}(Q, P)$ 
2:  $\mathcal{K} \leftarrow \mathcal{K}_Q \cup \mathcal{K}_P$ 
3: Initialize weight list  $\mathcal{W}$ 
4: for  $i = 1$  to  $|\mathcal{K}|$  do
5:    $w_i \leftarrow \text{GetWeight}(k_i, M)$ 
6:    $w_c \leftarrow \text{DefaultWeight}(k_i)$ 
7:    $w \leftarrow w_m + w_c$ 
8:   Add  $(k_i, w_i)$  to  $\mathcal{W}$ 
9: end for
10: Perform weighted keyword retrieval:
     $\mathcal{S}_{k1} \leftarrow \text{KeywordMatch}(\mathcal{K}, \mathcal{KB})$ 

11: Initialize document vector list  $\mathcal{V}$ 
12: for each document  $D_i$  in  $\mathcal{S}_{k1}$  do
13:    $\mathbf{v}_i \leftarrow \text{Vectorize}(D_i)$ 
14:   Add  $\mathbf{v}_i$  to  $\mathcal{V}$ 
15: end for
16:  $\mathbf{q} \leftarrow \text{Vectorize}(Q)$ 
17: Initialize similarity list  $\mathcal{S}_{sim}$ 
18: for  $j = 1$  to  $|\mathcal{V}|$  do
19:   Compute cosine similarity  $\text{Sim}_j$ 
20:   Add  $(D_j, \text{Sim}_j)$  to  $\mathcal{S}_{sim}$ 
21: end for
22: Sort  $\mathcal{S}_{sim}$  by similarity score
23: Select top- $k_2$  documents:  $\mathcal{D}_{top} \leftarrow \text{TopK}(\mathcal{S}_{sim}, k_2)$ 
24: Return  $\mathcal{D}_{top}$ 

```

to each keyword, forming a keyword list and weight list $M = \{(k_1, w_1), (k_2, w_2), \dots, (k_m, w_m)\}$, where k_i represents the keyword, and w_i represents the weight of the keyword k_i . Meanwhile, the user intent and deployable commands are processed using traditional tokenization methods, generating a keyword list $N = \{(k_1, w_c), (k_2, w_c), \dots, (k_n, w_c)\}$ with default weights w_c . Subsequently, the lists M and N are merged to form a new list L . The final weight is the sum of the weights from both lists:

$$w = w_i + w_c \quad (5)$$

Next, the list L is used as input to perform the weighted keyword retrieval algorithm:

$$S = \sum_{i=1}^{|Q|} \text{IDF}(Q_i) \cdot \frac{f(Q_i, D) \cdot w(Q_i) \cdot (k_3 + 1)}{f(Q_i, D) + w(Q_i) \cdot (k_3 + w(Q_i)) \cdot K} \quad (6)$$

The weighted score of the query S is determined by inverse document frequency $\text{IDF}(Q_i)$, adjustment parameter K , and term frequency $f(Q_i, D)$, with a term weight function $w(Q_i)$. Parameters include w for query weight.

This step retrieves documents related to a specific business scenario from the external network metric KB, computes a score for each document, and outputs the most relevant documents along with their scores.

Services Analysis Guide

The known user intent is: **<given_intnet>**. Based on the user intent, you need to analyze the weight allocation matrix of the following QoS indicators based on the characteristics: **<Indicator_1>**, **<Indicator_2>**, **<Indicator_3>**.

Output Format

```

{
  "Indicator_1": {
    "value": "<Indicator_value_1>",
    "reason": "<reason_1>"
  },
  "Indicator_2": {
    "value": "<Indicator_value_2>",
    "reason": "<reason_2>"
  }, ...
}

```

Fig. 5. The example prompt for service analysis by ITLLM.

2) *Vector Retrieval Stage*: In the initialization process, the system performs vectorization on all documents. This process converts the content of the documents into high-dimensional vector representations, allowing each document to have a unique representation in the vector space. The vector of each document not only captures the content information of the document but also reflects the semantic similarity between documents.

After the initial retrieval using the weighted keyword retrieval algorithm, the system obtains a series of document scores. Typically, documents with a score of 0 are considered irrelevant to the current query scenario, and therefore, their vector representations are removed from the retrieval index. This way, subsequent retrieval operations are not interfered with by irrelevant documents, making the retrieval process more efficient and accurate.

Next, the system focuses on processing the top- k_1 document vectors. The key operation in this stage is to compute the similarity between the query vector and the document vectors. The most commonly used method for calculating similarity is cosine similarity, which measures the angle between two vectors.

Based on these similarity metrics, the system calculates the similarity score between the query vector and each document vector. Finally, the system returns the top- k_2 documents with the highest scores as the most relevant retrieval results to the user, ensuring that the returned documents are not only semantically relevant to the query but also capable of efficiently handling large-scale document collections, providing accurate retrieval results.

D. Resource Allocator

The Resource Allocator primarily consists of ITLLM and a DRL model. ITLLM decomposes the services in the intent into a QoS weight matrix based on service characteristics, with a sample prompt shown in Fig. 5. ITLLM utilizes ChatGLM-32B to analyze the services contained in the intent. Then, the QoS weight allocation matrix is determined based on

the characteristics of each QoS. Subsequently, the matrix is incorporated into the input to the DRL model state, ultimately yielding the optimal resource value allocated to users. LLMs are trained on massive datasets, providing strong generalization and transfer capabilities, enabling them to handle the feature analysis of emerging services better.

A DRL model is trained to maximize the benefits of user services by selecting the appropriate bandwidth. On the current network, different user services include the following three QoS indicators: Data Transfer Rate (DTR), Packet Loss Rate (PLR), and Bit Error Rate (BER).

In the context of networking, users frequently demand a high DTR to guarantee the swift loading of high-definition videos. Additionally, a low PLR is essential to reduce the stuttering of live videos. Furthermore, a low BER is needed to lessen the probability of video and audio desynchronization. The Shannon formula is utilized to compute the DTR for each link, with the link's average value serving as the metric for calculating QoS benefits. This approach ensures a comprehensive evaluation of network performance, catering to the diverse needs of users:

$$\mathcal{O}_{DTR} = \frac{1}{L_{n_u, d_u}} \sum_{i \in L_{n_u, d_u}} B_u \log_2 \left(1 + \frac{P_i}{\sigma_i} \right) \quad (7)$$

where P_i is the signal power of the link i , and σ_i is the noise power of the link i .

The computation of the two QoS indicators, PLR and BER, is simulated considering a variety of influential factors. The upper incomplete gamma function is employed to simulate the PLR in response to alterations in bandwidth, noise power, and the number of links. The subsequent section outlines the methodology for calculating the benefits associated with PLR. This approach provides a comprehensive understanding of network performance under varying conditions:

$$\mathcal{O}_{PLR} = 1 - \frac{1}{L_{n_u, d_u}} \sum_{i \in L_{n_u, d_u}} \frac{\Gamma(L_{n_u, d_u}, \frac{B_u}{\sigma_i^2})}{\Gamma(L_{n_u, d_u})} \quad (8)$$

where $\Gamma(\cdot)$ represents the upper incomplete gamma function. In order to calculate the gain of BER, it is necessary to simulate the change in BER in relation to bandwidth, signal-to-noise ratio, and the maximum error rate of a single bit, which is denoted by g . The following outlines the method for calculating the gain of BER:

$$\mathcal{O}_{BER} = 1 - \frac{1}{L_{n_u, d_u}} \sum_{i \in L_{n_u, d_u}} (1 - g)^{\frac{\theta}{B_u \log_2 \left(1 + \frac{P_i}{\sigma_i} \right)}} \quad (9)$$

where θ is an empirical parameter, obtained by simulating the change in BER. By using the above three calculation formulas, we can effectively train a DRL model that maximizes the benefits of QoS by choosing the optimal bandwidth.

A Markov decision process is established in which the network determines bandwidth allocation decisions contingent on the present network state and user service. The state of the network S_u when the user service comes is composed of two parts: (1) the user service requirement can be represented by a five-element vector $\{n_u, d_u, W_u\}$, which includes the source node n_u , destination node d_u of the service request, and QoS

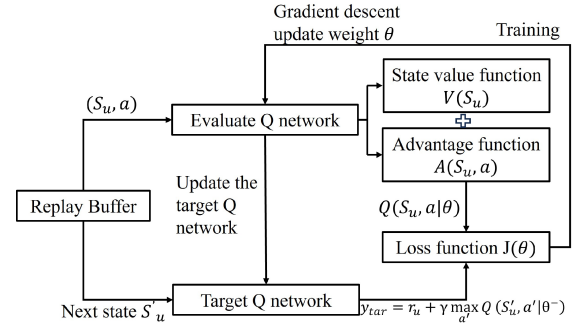


Fig. 6. The workflow of the DuelingDQN training process for resource allocation.

weight allocation matrix $W_u = \{w_{u, \mathcal{O}_{DTR}}, w_{u, \mathcal{O}_{PLR}}, w_{u, \mathcal{O}_{BER}}\}$; (2) the network resource state is a depiction of the network state, encompassing the available bandwidth capacity of each network link $\{B_l : l \in L\}$. The action space, $\mathcal{A} = B_a = \{b_1, b_2, \dots, b_a\}$, consists of the bandwidths that the network links can set. The rewards, denoted by $r_u = R_u$, are the total QoS benefits for each service obtained from the interaction between the model and the environment based on the Eq. (3).

The DQN algorithm typically solves discrete problems like bandwidth resource allocation. However, the DQN algorithm employs the same values to both select and evaluate an action, which results in excessively optimistic value predictions. Furthermore, when confronted with a large state space and action space, the DQN algorithm exhibits low sampling efficiency and unstable training, rendering it challenging to obtain optimal policies [42]. To address the above issues, we employ the Dueling DQN.

Fig. 6 depicts the Dueling DQN training procedure. Dueling DQN is built upon DQN, jointly utilizing the Dueling network and Double DQN. The Dueling network divides the abstract features of the fully connected layer into two branches. One represents the state value function $V(S_u)$, which is the conditional expectation of the cumulative reward function with respect to the state. The other is the advantage function $A(S_u, a)$, which is the additional value brought by selecting an action. The sum of the two yields the value $Q(S_u, a)$, which indicates the value of taking a specific action a from state S_u , and the network executes the action that obtains the maximum $Q(S_u, a)$ [43].

The model's parameters are denoted as $\theta = (\delta, \eta, \rho)$. The DuelingDQN model typically has a shared feature extraction layer δ , which consists of multiple fully connected layers. After feature extraction, the network splits into two branches. One branch is used to compute the state-value function $V(S_u)$. This branch uses fully connected layers η to learn the function representing the state value. The other branch is used to compute the advantage function $A(S_u, a)$. Similarly, it uses fully connected layers ρ to learn the advantage function [43].

Throughout the training phase, the replay buffer D is employed to accumulate the experience samples (S_u, a, r_u, S_u') in the model. The parameters θ and θ^- correspond to the evaluation Q network and target Q network in the model, respectively. In order to facilitate the model training, a random

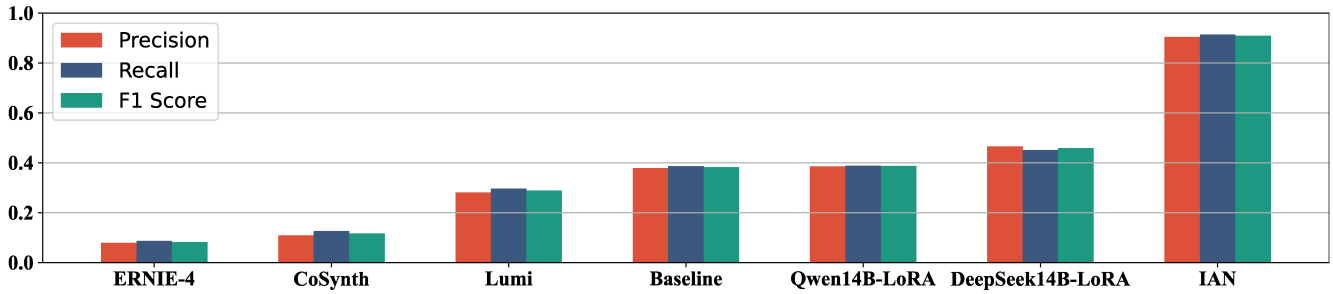


Fig. 7. The comparison of the Precision, Recall, and F1 Score results for intent translation between ERNIE-4, CoSynth, Lumi, ChatGLM-6B (baseline), Baichuan-13B-LoRA, DeepSeek14B-LoRA and IAN.

sample is extracted from the replay buffer with the objective of minimizing the loss. The DuelingDQN algorithm is used for model training to choose the optimal action for maximizing the total QoS benefits of user service.

V. EXPERIMENTS

A. Experimental Settings

1) *Simulated Network Setting*: We utilize the INS IXC service network topology, the IRIS network, and the Interoute network [44] as the three foundational network topologies of different scales for our simulations. The INS IXC comprises 33 nodes and 41 links. The IRIS network consists of 51 nodes and 64 links, and the Interoute has 110 nodes and 149 links. Of these topologies, seven, fifteen, and twenty-four are designated for user equipment, while the remaining nodes function as base stations and switches within the network. Each link is initially configured with bandwidth resources of 100 Mbps, a signal power of 1000W, and a noise power of 1W. We introduce a parameter, denoted as ξ , to represent the network resource availability coefficient. The value of ξ varies from 0.3 to 1. For example, when $\xi = 0.5$, the maximum capacity of the links is reduced to 50 Mbps. Conversely, when $\xi = 1.0$, the links can operate at their full capacity of 100 Mbps. This parameter allows us to simulate various network conditions and analyze their impact on performance. We used services and service weights that were not previously utilized in DRL training when testing resource allocation, with a new-to-old ratio of 3:7. Meanwhile, the interval at which the intent of the service is generated is randomly selected from a range of values between [1, 5] seconds.

2) *Relevant Parameter Setting*: In the realm of intent translation, we amass a collection of open-source network management commands and manuals to serve as both datasets and KB. We utilize the ChatGLM-6B model for full-parameter fine-tuning training, employing the Adam optimizer with a learning rate set to 10^{-4} . The dataset is partitioned into training and testing sets at a ratio of 10:1. Subsequently, we continued to utilize the ChatGLM-6B to complete the remaining tasks associated with the RAG project. For resource allocation, the parameters are configured as follows: $\beta_1 = 0.7$, $\beta_2 = 0.3$, $\alpha_1 = \alpha_2 = 0.5$, $r = 10$, $\vartheta = 5000$, and $g = 0.01$. Dueling DQN is trained using the Adam optimizer with a learning rate of 10^{-3} and $\gamma = 0.1$, a batch size of 32, for 1000 epochs. The interval for transferring parameters between the estimated Q

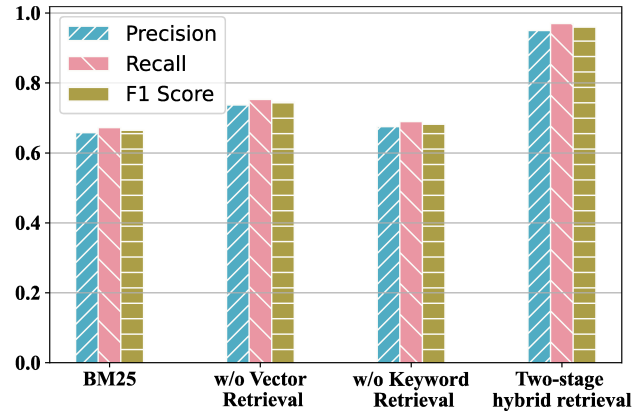


Fig. 8. Results of ablation experiments for two-stage hybrid retrieval.

network and the target Q network is set to 1000. The selection of nodes for the user and server is contingent upon the user's intent, and the shortest path is computed to determine the base stations traversed between them.

B. Performance of Intent Translator

To demonstrate the effectiveness of the Intent Translator, we conducted comparative experiments using ERNIE-4, CoSynth, Lumi, ChatGLM-6B (baseline), Qwen-14B-LoRA [45], DeepSeek14B-LoRA and IAN. The results are illustrated in the Fig. 7. ChatGLM-6B, Qwen-14B and DeepSeek-14B adopt an end-to-end fine-tuning approach for intent translation tasks. Since ERNIE-4 does not provide an interface for fine-tuning, we prompt it with instructions. For CoSynth, only its intent translation component was tested. For Lumi, we trained its LSTM model end-to-end and only evaluated whether the generated command parameters met the requirements. The experimental results show that the precision, recall, and F1 score of ERNIE-4 are 7.89%, 8.65% and 8.25% respectively. This result indicates that although commercial LLMs achieve good results in general domains, they underperform in the tasks like intent translation. Moreover, compared to the baseline, IAN showed improvements in the precision, recall, and F1 score by 51.83%, 53.49%, and 52.66% each compared to baseline, which can be attributed to the effects of step 1 and step 3 shown in Fig. 3. To further substantiate this, we designed ablation experiments, as seen in Table II. After adding step 1, there was an increase in the F1 score by 12.27%

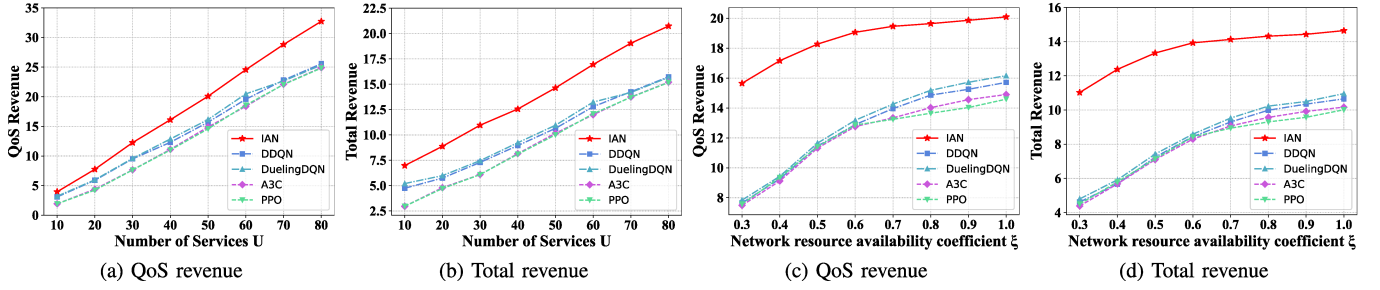


Fig. 9. The performance of IAN in comparison to other DRL models with unchanged service QoS weight allocation under different numbers of services U and varying network resource availability coefficients ξ under the network topology of INS IXC.

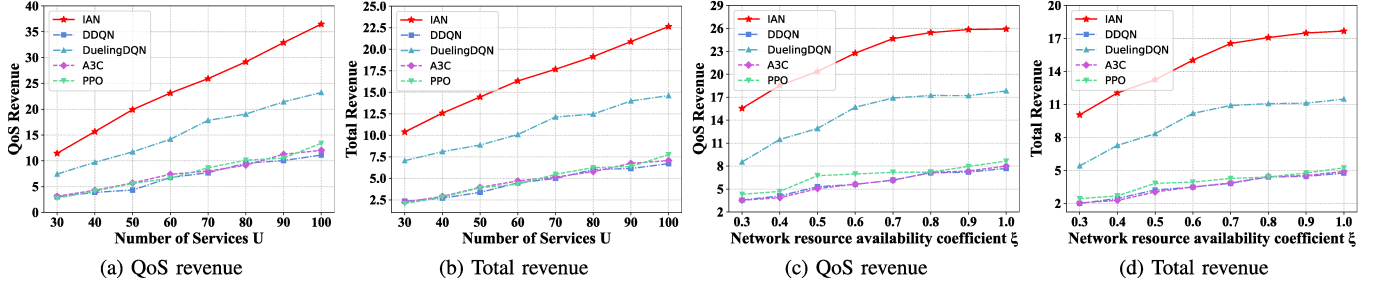


Fig. 10. The performance of IAN in comparison to other DRL models with unchanged service QoS weight allocation under different numbers of services U and varying network resource availability coefficients ξ under the network topology of IRIS.

TABLE II

THE PRECISION, RECALL, AND F1 SCORE OF THE INTENT TRANSLATION TASK WERE COMPARED BEFORE AND AFTER THE INTRODUCTION OF STEPS 1 AND 3

Model	Metric		
	Precision	Recall	F1 score
Baseline	38.62%	37.88%	38.25%
Baseline+step 1	50.82%	50.23%	50.52%
Baseline+step 3	76.23%	73.42%	74.80%
IAN	90.45%	91.37%	90.91%

compared to the baseline. Adding step 3 resulted in an F1 score improvement of 36.55% compared to the baseline.

C. Performance of Two-Stage Hybrid Retriever

To evaluate the performance of the two-stage hybrid retrieval algorithm for related document retrieval based on user intent, we designed an ablation study. This involved assessing the retrieval effectiveness of keyword retrieval and vector retrieval both independently and in combination for 60 intents, using the traditional BM25 [46] algorithm as a baseline. Precision, Recall, and F1 Score were used to quantify the retrieval effectiveness. As showed in Fig. 8, the experimental conclusion indicates that the two-stage hybrid retrieval algorithm improves the F1 Score by 44.3%, 28.8%, and 40.7% compared to BM25, keyword retrieval, and vector retrieval respectively. This demonstrates that the two-stage hybrid retrieval algorithm has significant advantages, as it enhances recall while maintaining high precision, thereby improving overall retrieval performance.

D. Performance of Resource Allocator

To prove the effectiveness of IAN in resource allocation, we designed and constructed a simulation network based on three different scales of real network topologies and simulated real business scenarios (see Section V-A for details). We evaluated the performance of IAN using two metrics: 1) QoS Revenue, which measures the performance of IAN considering only QoS indicators, is calculated as the sum of all service QoS benefits according to Eq. (4a). 2) Total Revenue, which measures the performance of IAN considering both QoS and SSR, is calculated as the total revenue from all services according to Eq. (3). Additionally, we selected the following four DRL models for the comparative experiments:

- **DDQN** [48] Uses two networks to separate action selection and evaluation, reducing overestimation issues and improving training stability and performance.
- **DuelingDQN** Introduces separate branches for the state value function and action advantage function, better-evaluating state values.
- **A3C** [49] Runs multiple agents in parallel and asynchronously updates global network parameters, enhancing training efficiency and stability.
- **PPO** [50] Limits the extent of policy updates to maintain training stability and sample efficiency, excelling in various tasks.

A comparative experiment was conducted with IAN compared to the above DRL models across three different scales of network topologies, while keeping the service QoS weight allocation unchanged throughout the experiment. The purpose of this experiment is to validate whether the guidance of LLM is helpful for the training and inference of DRL models in resource allocation tasks, and to evaluate how IAN performs

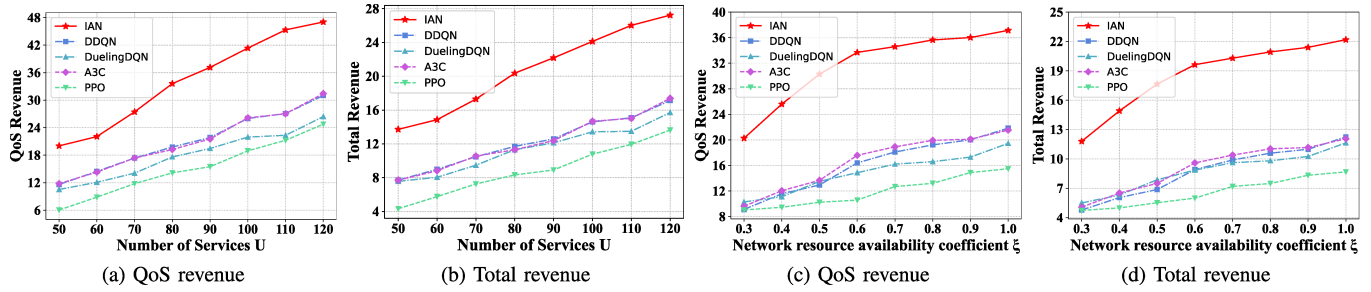


Fig. 11. The performance of IAN in comparison to other DRL models with unchanged service QoS weight allocation under different numbers of services U and varying network resource availability coefficients ξ under the network topology of Interoute.

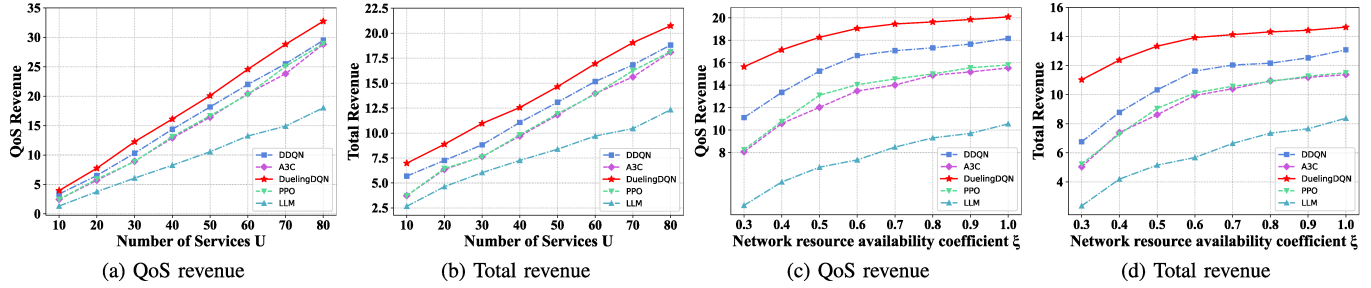


Fig. 12. The performance of different DRL models under different numbers of services U and varying network resource availability coefficients ξ .

in larger and more complex network environments. The results are presented in Fig. 9, Fig. 10, and Fig. 11. The simulation network emulated scenarios with varying numbers of services and resource scarcity to showcase the performance of different models. According to Fig. 9(a), and Fig. 9(b), when the number of services increased to 80, IAN improved QoS revenue and total revenue by 27.99% and 31.85% respectively, compared to the suboptimal method. Similarly, under the network resource availability coefficient $\xi = 0.3$, Fig. 9(c) and Fig. 9(d) show that IAN outperformed the suboptimal method in QoS revenue and total revenue by 24.31% and 33.57%. In the comparisons with other topologies, it is also shown that IAN has the best performance and is least affected by different network scales. The experimental results demonstrate that the utilization of LLM for QoS weight allocation allows for a more reasonable distribution of resources for user services. IAN continues to maintain a strong performance in larger and more complex network environments, demonstrating its robust scalability.

The introduction of LLM for analyzing new services and generating QoS metric weights based on service characteristics guides the training and inference of models. Therefore, compared to other DRL models, IAN better adapts to the emergence of new services and demonstrates stronger robustness. Additionally, we designed ablation experiments to compare the performance of different DRL models when integrating LLM, and the performance of LLM alone in resource allocation, as shown in Fig. 12. The experiments prove that DuelingDQN, compared to A3C and PPO, possesses superior capabilities in handling discrete data and is better suited for resource allocation tasks. Additionally, the experiment also shows that combining LLM with DRL can effectively ensure that user demands are met. In contrast, using LLM alone for resource

allocation is prone to being affected by hallucinations, leading to suboptimal allocation results.

VI. CONCLUSION

To address the challenges faced by AN, we propose IAN, an intent-based AN framework guided by LLM. IAN leverages LLMs to produce high-quality commands under complex constraints, ensuring robust and adaptable network management. This integration allows for a more reasonable resource distribution, meeting the diverse demands of emerging services. IAN is comprised of two primary components: the Intent Translator and the Resource Allocator. The Intent Translator converts user intent into deployable commands using RAG to improve the quality of commands. The Resource Allocator uses LLMs to analyze service characteristics and guide resource distribution effectively. The superior performance of IAN in accurate demand comprehension and resource allocation highlights its potential as a significant advancement in intelligent network management solutions. This capability ensures optimal utilization of network resources, thereby enhancing overall efficiency and user satisfaction.

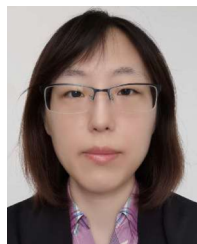
REFERENCES

- [1] A. Shahraki, M. Abbasi, M. Jalil Piran, and A. Taherkordi, "A comprehensive survey on 6G networks: Applications, core services, enabling technologies, and future challenges," 2021, *arXiv:2101.12475*.
- [2] H. F. Alhashimi et al., "A survey on resource management for 6G heterogeneous networks: Current research, future trends, and challenges," *Electronics*, vol. 12, no. 3, p. 647, Jan. 2023.
- [3] Z. Zhang et al., "6G wireless networks: Vision, requirements, architecture, and key technologies," *IEEE Veh. Technol. Mag.*, vol. 14, no. 3, pp. 28–41, Sep. 2019.
- [4] S. Dang, O. Amin, B. Shihada, and M.-S. Alouini, "What should 6G be?," *Nature Electron.*, vol. 3, no. 1, pp. 20–29, Jan. 2020.

- [5] A. Richard et al., "Autonomous networks: Empowering digital transformation for telecoms industry," TM Forum, Tech. Rep., 2019.
- [6] M. Behringer et al., "Autonomic networking: Definitions and design goals," Internet Res. Task Force, Tech. Rep. rfc7575, 2015.
- [7] B. Min et al., "Recent advances in natural language processing via large pre-trained language models: A survey," *ACM Comput. Surveys*, vol. 56, no. 2, pp. 1–40, Feb. 2024.
- [8] W. Xin Zhao et al., "A survey of large language models," 2023, *arXiv:2303.18223*.
- [9] K. Mehmood, K. Kravetska, and D. Palma, "Intent-driven autonomous network and service management in future cellular networks: A structured literature review," *Comput. Netw.*, vol. 220, Jan. 2023, Art. no. 109477.
- [10] R. Chaparadza, M. Elkotob, B. Radier, T. B. Meriem, E. Hammad, and T. Choi, "Autonomic/autonomous networking (AN): Federations & governance of ANs, progress & open challenges in standards: Analysis report," in *Proc. IEEE Future Netw. World Forum (FNWF)*, Oct. 2022, pp. 176–183.
- [11] A. S. Jacobs et al., "Hey, lumi! Using natural language for intent-based network management," in *Proc. USENIX Annu. Tech. Conf.*, 2021, pp. 625–639.
- [12] A. S. Jacobs, R. J. Pfitscher, R. A. Ferreira, and L. Z. Granville, "Refining network intents for self-driving networks," in *Proc. Afternoon Workshop Self-Driving Netw.*, Aug. 2018, pp. 15–21.
- [13] T. Szyrkowiec et al., "Automatic intent-based secure service creation through a multilayer SDN network orchestration," *J. Opt. Commun. Netw.*, vol. 10, no. 4, pp. 289–297, Apr. 2018.
- [14] A. Mekrache and A. Ksentini, "LLM-enabled intent-driven service configuration for next generation networks," in *Proc. IEEE 10th Int. Conf. Netw. Softwarization (NetSoft)*, Jun. 2024, pp. 253–257.
- [15] R. Mondal, A. Tang, R. Beckett, T. Millstein, and G. Varghese, "What do LLMs need to synthesize correct router configurations?," in *Proc. 22nd ACM Workshop Hot Topics Netw.*, Nov. 2023, pp. 189–195.
- [16] J. Achiam et al., "GPT-4 technical report," 2023, *arXiv:2303.08774*.
- [17] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," 2023, *arXiv:2307.09288*.
- [18] A. Zeng et al., "GLM-130B: An open bilingual pre-trained model," 2022, *arXiv:2210.02414*.
- [19] Y. Sun et al., "ERNIE 3.0: Large-scale knowledge enhanced pre-training for language understanding and generation," 2021, *arXiv:2107.02137*.
- [20] A. Liu et al., "DeepSeek-V3 technical report," 2024, *arXiv:2412.19437*.
- [21] A. Yang et al., "Qwen3 technical report," 2025, *arXiv:2505.09388*.
- [22] L. Hu, Z. Liu, Z. Zhao, L. Hou, L. Nie, and J. Li, "A survey of knowledge enhanced pre-trained language models," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 4, pp. 1413–1430, Apr. 2024.
- [23] J. Yang et al., "Harnessing the power of LLMs in practice: A survey on ChatGPT and beyond," *ACM Trans. Knowl. Discovery from Data*, vol. 18, no. 6, pp. 1–32, Jul. 2024.
- [24] W. Chen et al., "LLM-enabled incremental learning framework for hand exoskeleton control," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 2617–2626, 2024.
- [25] Y. Zhang, Y. Cao, X. Xu, and W. Shen, "LogiCode: An LLM-driven framework for logical anomaly detection," *IEEE Trans. Autom. Sci. Eng.*, vol. 22, pp. 7712–7723, 2025.
- [26] H. Xu, Y. Jin Kim, A. Sharaf, and H. Hassan Awadalla, "A paradigm shift in machine translation: Boosting translation performance of large language models," 2023, *arXiv:2309.11674*.
- [27] J. Guo, H. Yang, Z. Li, D. Wei, H. Shang, and X. Chen, "A novel paradigm boosting translation capabilities of large language models," 2024, *arXiv:2403.11430*.
- [28] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9459–9474.
- [29] S. Xu et al., "Unsupervised information refinement training of large language models for retrieval-augmented generation," 2024, *arXiv:2402.18150*.
- [30] H. Qian, Z. Liu, K. Mao, Y. Zhou, and Z. Dou, "Grounding language model with chunking-free in-context retrieval," 2024, *arXiv:2402.09760*.
- [31] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," 2023, *arXiv:2312.10997*.
- [32] Z. Li et al., "PET-SQL: A prompt-enhanced two-round refinement of text-to-SQL with cross-consistency," 2024, *arXiv:2403.09732*.
- [33] Y. Kong et al., "TPTU-v2: Boosting task planning and tool usage of large language model-based agents in real-world systems," 2023, *arXiv:2311.11315*.
- [34] A. Singh, A. Ehtesham, S. Kumar, and T. Talaei Khoei, "Agentic retrieval-augmented generation: A survey on agentic RAG," 2025, *arXiv:2501.09136*.
- [35] S.-Q. Yan, J.-C. Gu, Y. Zhu, and Z.-H. Ling, "Corrective retrieval augmented generation," *CoRR*, vol. abs/2401.15884, 2024.
- [36] S. Jeong, J. Baek, S. Cho, S. Ju Hwang, and J. C. Park, "Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity," 2024, *arXiv:2403.14403*.
- [37] N. C. Luong et al., "Applications of deep reinforcement learning in communications and networking: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3133–3174, 4th Quart., 2019.
- [38] F. Mason, G. Nencioni, and A. Zanella, "Using distributed reinforcement learning for resource orchestration in a network slicing scenario," *IEEE/ACM Trans. Netw.*, vol. 31, no. 1, pp. 88–102, Feb. 2023.
- [39] Y. Liu et al., "Slice sandwich: Jagged slicing multi-tier dynamic resources for diversified V2X services," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4285–4302, May 2024.
- [40] R. Han et al., "Multi-SP network slicing parallel relieving edge network conflict," *IEEE Trans. Parallel Distrib. Syst.*, vol. 34, no. 11, pp. 2860–2875, Nov. 2023.
- [41] T. Dong et al., "Intelligent joint network slicing and routing via GCN-powered multi-task deep reinforcement learning," *IEEE Trans. Cognit. Commun. Netw.*, vol. 8, no. 2, pp. 1269–1286, Jun. 2022.
- [42] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "Deep reinforcement learning: A brief survey," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, Nov. 2017.
- [43] Z. Wang, T. Schaul, M. Hessel, H. van Hasselt, M. Lanctot, and N. de Freitas, "Dueling network architectures for deep reinforcement learning," 2015, *arXiv:1511.06581*.
- [44] S. Knight, H. X. Nguyen, N. Falkner, R. Bowden, and M. Roughan, "The internet topology zoo," *IEEE J. Sel. Areas Commun.*, vol. 29, no. 9, pp. 1765–1775, Oct. 2011.
- [45] J. Bai et al., "Qwen technical report," 2023, *arXiv:2309.16609*.
- [46] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Found. Trends Inf. Retr.*, vol. 3, no. 4, pp. 333–389, 2009.
- [47] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Text Summarization Branches Out*, 2004, pp. 74–81.
- [48] H. V. Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," *AAAI*, vol. 30, no. 1, pp. 2094–2100, Mar. 2016.
- [49] V. Mnih et al., "Asynchronous methods for deep reinforcement learning," in *Proc. 33rd Int. Conf. Mach. Learn.*, vol. 48, 2016, pp. 1928–1937.
- [50] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.



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